

Gabriel MARIADASS | Paris, France | French

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SUMMARY

M.Sc. student in Mathematics, Vision, and Machine Learning (MVA) at ENS Paris-Saclay, currently researching possibilistic variational inference at NTU Singapore. Pairs that research with applied ML engineering experience (Dassault Systèmes, Banque de France). Looking to bring both to an AI startup or scale-up building real products.

EDUCATION

ENS Paris-Saclay, France

2025 - 2026

M.Sc. Mathematics, Vision, and Machine Learning (MVA)

Academic average: 17.01/20 (4.0 GPA)

Key courses: Bayesian Machine Learning, Deep Learning, Graph in ML, Advanced Learning for Text and Graph Data, Computational Statistics, Interactions, Statistical Learning with Extreme Values

École Nationale des Ponts et Chaussées, France

2022 - 2026

Master's Year 1 - Mathematical and Computer Engineering (IMI)

Lycée Henri-IV, France

2020 - 2022

CPGE - PCSI/PC*

RESEARCH EXPERIENCE

LPSM, France & NTU, Singapore

04/2026 - 08/2026

Research Intern - Possibilistic Variational Inference

Supervisors: Badr-Eddine Chérif-Abdellatif (CNRS, LPSM) and Jérémie Housseineau (NTU)

- Wrote foundational notes on uncertainty, variational inference, information, and possibility theory
- Studied the Maxitive Donsker-Varadhan formulation, replacing expectation with supremum-based optimization
- Extended the framework with four anisotropic precision structures, proving a Picard-type convergence theorem
- Connected the optimizer to Sharpness-Aware Minimization via the Bayesian Learning Rule

PROFESSIONAL EXPERIENCE

Dassault Systèmes, France

01/2025 - 07/2025 (Second part of gap year)

ML Engineer - Deep Learning & GNN (Molecular Retrosynthesis)

- Improved ML pipeline scalability and maintainability
- Applied PyTorch Geometric for graph-based molecular modeling
- Used model quantization and Bayesian hyperparameter optimization

Banque de France, France

09/2024 - 12/2024 (First part of gap year)

Data Scientist - On-site Credit Institution Control Delegation

- Developed credit risk ML models on large financial datasets
- Performed statistical and econometric analysis on time series data

TECHNICAL SKILLS

- **Mathematics:** Probability, Statistics, Convex Optimization, Stochastic Processes, Possibility Theory
- **ML:** PyTorch, PyTorch Geometric, GNNs, Variational Inference, PAC-Bayes, Hyperparameter Optimization, Quantization
- **Programming:** Python, C++, SQL, R, SAS
- **Tools:** Git, VSCode, Jupyter, Overleaf

LANGUAGES, SPORTS & INTERESTS

- **Languages:** French (native), English (C1), Spanish (B1)
- **Interests:** Wellbeing, Muay Thai, Fitness, Spirituality, History, Travel, Research, Machine Learning, AI, Entrepreneurship